

Topology MATM36: Course Summary of Important Results

Simon Gustafsson

Baire Category Theorem

Theorem 1 (Baire Category Theorem). *Let (X, d) be a complete metric space and let $\{U_n\}_{n=1}^\infty$ be a sequence of dense open subsets of X . Then $\bigcap_{n=1}^\infty U_n$ is dense in X .*

Proof. To be added. □

Corollary 1. *Let (X, d) be a complete metric space and let $\{F_n\}_{n=1}^\infty$ be a sequence of nowhere dense subsets of X . Then*

$$\text{int}\left(\bigcup_{n=1}^\infty F_n\right) = \emptyset.$$

Proof. Since F_n is nowhere dense, $U_n := X \setminus \overline{F_n}$ is open and dense; by Baire, $\bigcap_{n=1}^\infty U_n$ is dense, hence nonempty in every nonempty open set, i.e. $\bigcup_{n=1}^\infty \overline{F_n}$ (and thus $\bigcup_{n=1}^\infty F_n$) has empty interior. □